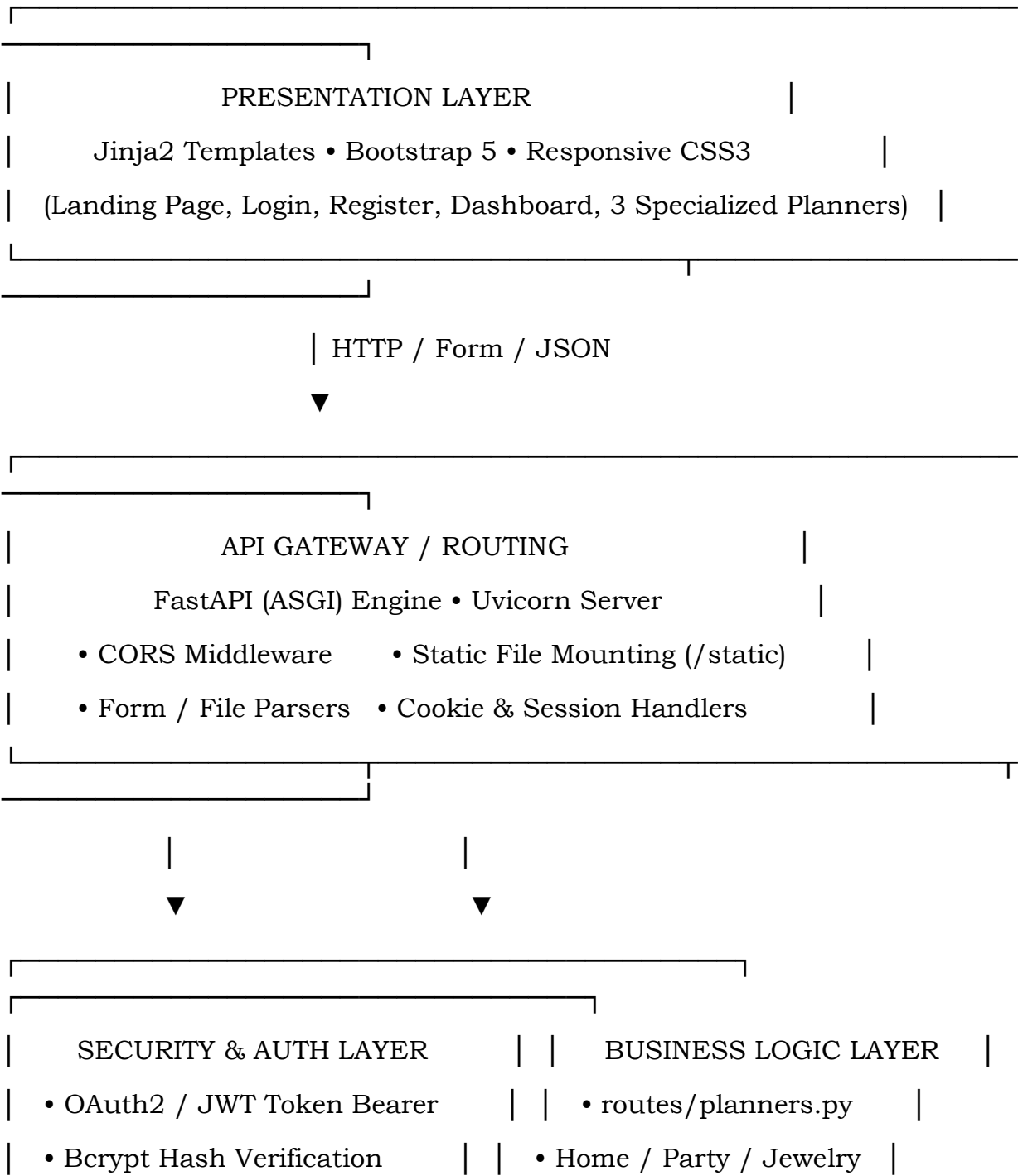
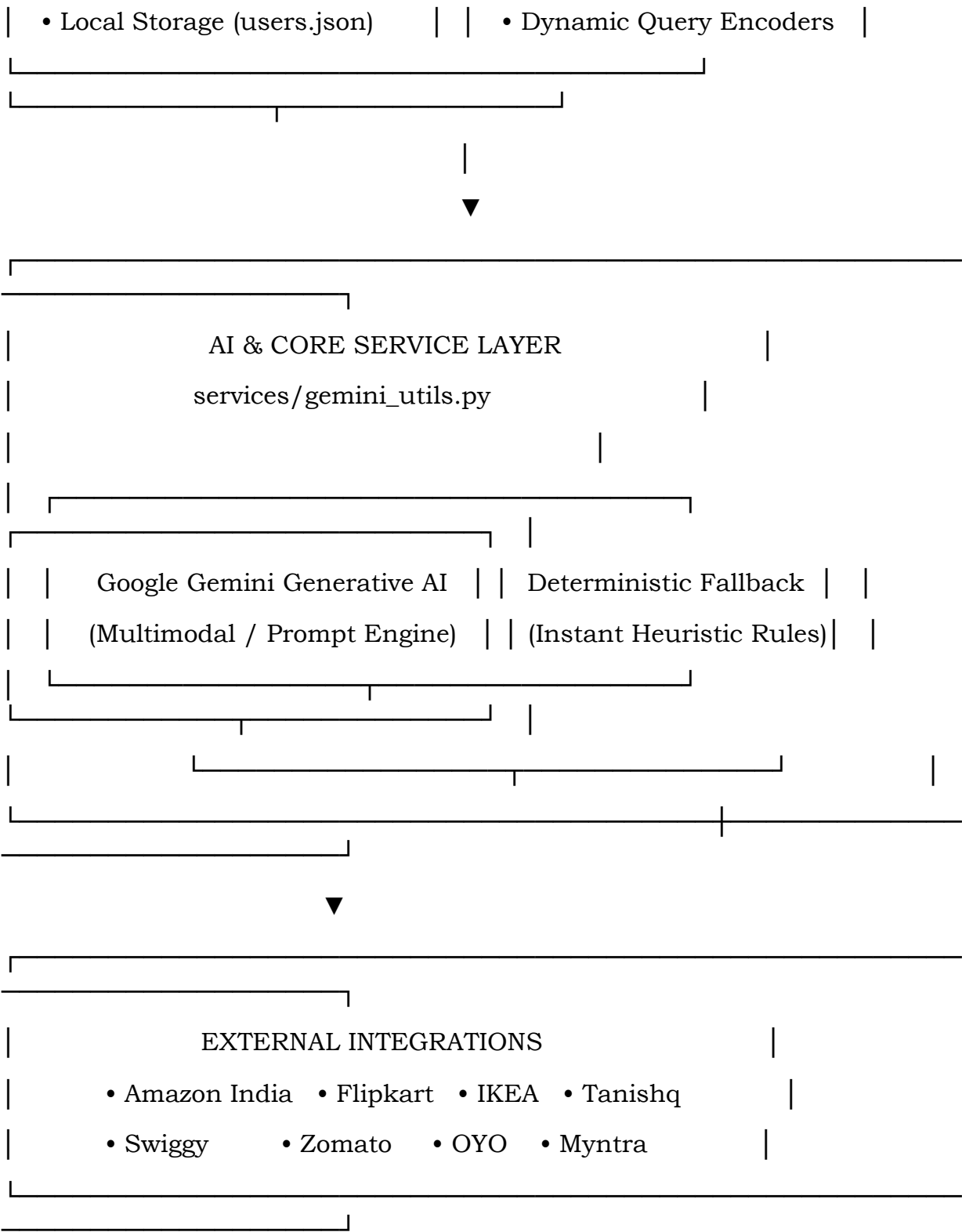


POCKETSMART AI: SOLUTION ARCHITECTURE & DESIGN

1. High-Level Architectural Diagram





2. Component Breakdown

2.1 Presentation Tier

- Server-side rendered views built with Jinja2 templates and Bootstrap 5.
- Clean, modular templates (index.html, dashboard.html, home_planner.html, party_planner.html, jewelry_planner.html).
- Asynchronous client fetch scripts for seamless form submission and dynamic card rendering.

2.2 Application Routing & Middleware

- FastAPI asynchronous ASGI router (routes/planners.py) separating domain controllers.
- Native file uploading pipeline storing temporary jewelry images in static/uploads/.
- Unified URL-encoded query parser (urllib.parse) generating accurate marketplace search endpoints.

2.3 Intelligent Processing Tier

- **Primary Path:** Prompt engineering framework instructing Gemini to output clean JSON schemas matching category metrics.
- **Secondary Resilient Path:** Built-in calculation module guaranteeing valid JSON responses even during network drops or zero-credit API conditions.

2.4 Data & Security Tier

- JWT signed access tokens (HS256 encryption) stored securely via browser cookies and headers.
- Local flat-file persistence (users.json) guaranteeing fast read/write operations without requiring external database servers.